

Matthew C. Voynovich

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EDUCATION

RENSSELAER POLYTECHNIC INSTITUTE

M.S. in Computer Science

Concentration in AI, Machine Learning, and Data Science

Cumulative GPA: 4.0/4.0

Troy, NY

Jan 2026 - May 2027

RENSSELAER POLYTECHNIC INSTITUTE

Dual Major B.S. in Computer Science and B.S. Information Technology & Web Science

Concentration in AI, Machine Learning, and Data Science

Cumulative GPA: 3.97/4.0; Dean's List 2023-2026; Accelerated to Graduate in 3 years; Summa Cum Laude

Relevant Coursework: Managing IT Resources, Algorithms, Database Systems, Web Science Development, Computational Vision, Intro to Logic Based AI, Machine Learning from Data, Data Science, Decentralized AI, Projects in AI & ML

Awards and Leadership: 6x Dean's Honor List; President of the Rensselaer Running Club, UPE Honor Society

Troy, NY

Aug 2023 - May 2026

WORK EXPERIENCE

IT/SECURITY/AUDIOVISUAL DESIGN INTERNSHIP

MGE UTC

New York, NY

June 2025 – Aug 2025

- Assisted in design and documentation of IT infrastructure, integrated security, and audiovisual systems for commercial projects.
- Produced technical design documentation in CAD & Revit, including floor plans, reflected ceiling plans, enlarged plans, elevations, riser diagrams, and construction details across IT, security, and audiovisual disciplines. Developed and sketched IDF closet layouts, coordinating telecommunications equipment, data connections, and rack configurations to support project requirements.
- Developed a Dynamo script for Revit to automate sheet list creation and track revision history across project sheets, improving documentation accuracy and workflow efficiency.

FULL STACK DEVELOPER

Contract Work for Novella Prep

Remote

June 2025 – Aug 2025

- Spearheaded the development of Zeewa, an adaptive learning platform featuring AI-generated practice, diagnostic exams, interactive lessons, and assignment workflows, launching a new product line and revenue stream for the test prep company.
- Built a scalable Vue.js + Node.js + PostgreSQL (Supabase) system using TypeScript, Docker, and Nginx. Integrated OpenAI LLMs for dynamic content creation and implemented real-time LaTeX/Markdown rendering, SSE pipelines, and stateful authentication. Managed full deployment workflow with DigitalOcean, Render, and Porkbun

RESEARCH

AI AGENT SYSTEMS & WORKFLOW SAFETY

Graduate Researcher, RPI

Troy, NY

Jan 2025 – Present

- Researching policy-compliant AI agent systems for dynamic, multi-step workflows, focusing on detecting and preventing privacy risks, sensitive information leakage, and policy violations across interactions with users, tools, and external data.
- Developing hybrid neuro-symbolic approaches that combine neural AI with symbolic reasoning and logic solvers to dynamically enforce policies, verify constraints, and adapt safeguards throughout evolving agent workflows.

OPTIMIZATION IN QUANTUM

Undergraduate Researcher, RPI

Troy, NY

Jun 2025 – Dec 2025

- Researched quantum computing applications in optimization by implementing QAOA and hybrid-classical optimization techniques in JAX and Qiskit to solve seismic inversion problems and compared quantum and classical machine learning approaches.

INVESTIGATIONS OF QUANTUM PHASE ESTIMATION

Undergraduate Researcher, RPI

Troy, NY

Jan 2025 – Jun 2025

- Researching applications of Quantum Phase Estimation (QPE) in Quantum Signal Processing. Explored mathematical foundations of quantum Fourier transforms, eigenvalue estimation, and phase estimation techniques to improve quantum signal processing.

PROJECTS

DECENTRALIZED MULTI-AGENT RL TRAFFIC CONTROL

Developer

Troy, NY

Apr 2026 – May 2026

- Developed a decentralized multi-agent reinforcement learning traffic control system in PyTorch using Deep Q-Networks (DQN) and a parameter-sharing CTDE (Centralized Training with Decentralized Execution) architecture to optimize traffic flow and stabilize scalable learning across 27 intersections in downtown Troy, NY.
- Achieved significant performance improvements over traditional traffic control strategies, reducing average vehicle waiting time from ~36 seconds to ~2 seconds in simulation and reducing queue lengths by ~6x.

ENTERPRISE SEARCH AGENT

Independent Contractor

Troy, NY

Feb 2026 – May 2026

- Built an AI-powered enterprise search prototype for Deloitte as part of a capstone project using Next.js (TypeScript), FastAPI, and PostgreSQL; implemented document ingestion, chunking, and embedding generation pipelines, then combined semantic vector search (pgvector) with BM25 keyword retrieval (ParadeDB) and reranking to deliver more relevant natural-language results.
- Designed the platform as an open-source, self-hosted solution to keep sensitive enterprise data fully onsite, reducing third-party exposure and strengthening security/compliance posture.

TECHNICAL SKILLS

Languages: Python, Java, C/C++, JavaScript
Web: HTML/CSS, React, Vue, Node.js, Express

Databases: SQL, MongoDB
ML/AI: Scikit-learn, PyTorch